

USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Title: Gen 4 CLx Exit in combination with Recovery Enabled timeout

Applied to: USB4 Specification Version 2.0

Brief description of the functional changes:

Gen 4 exit from CLx, in combination with Recovery enable, requires multiple timers to run in a serial manner. The first is on the exit from CL1/2 and then a new one in the Training state. This ECN simplifies the exit flow and fixes the Timeout Error and parameters definition.
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Benefits as a result of the changes:

Flow simplification

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
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NA

An analysis of the hardware implications:
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NA

An analysis of the software implications:
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NA

An analysis of the compliance testing implications:
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The requirements that are presented in this ECN are not backward compatible with the current specification definition. A device that already implemented the current definition will get a waiver for the new definition compliance testing.
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Actual Change

(a). Section 4.2.1.3.2.2 Gen 4 Behavior

Table4-34. Gen 4 Training Sub-State Machine Transitions

4	TS4	CL0	• Sent Gen 4 TS4 with the <i>Counter</i> field set to Fh.
5	Any	CLd	• Adapter remains in Training state for tTrainingAbort1 or tTrainingAbort2 time.
Notes: 1. All conditions need to be met before a transition takes place. 1-2. For a Gen 4 Link that has the Enable Gen 4 Link Recovery bit in PORT_CS_19 set to 0b, if the Adapter transitions from CLx state to Training state the count starts from the CLx Exit event (see Section 4.2.1.6.5.4) and not from the entering to Training state.			

(b). Section 4.2.1.3.3 Exit from State

To Text:

A Lane Adapter that transitions from a state other than CLd to Training state shall complete training and transition to the CL0 state within tTrainingAbort2 after entering the Training state. For a Gen 4 Link that has the Enable Gen 4 Link Recovery bit in PORT_CS_19 set to 0b, if the Adapter transitions from CLx state to Training state the count starts from the CLx Exit event (see Section 4.2.1.6.5.4) and not from the entering to Training state. If the Adapter does not transition to CL0 state within tTrainingAbort2, the Router shall initiate a Disconnect by driving SBTx to a logical low state for tDisconnectTx.

For a Gen 4 Link that has the Enable Gen 4 Link Recovery bit in PORT_CS_19 set to 1b, if the Adapter transitions from CLx state to Training state and does not transition to CL0 state within tTrainingError starting from the CLx Exit event (see Section 4.2.1.6.5.4), the Router shall initiate Gen 4 Link Recovery as described in Section 4.4.7.

(c). Section 4.2.1.6.5.1.2 Gen 4 CL0s Exit Flow

To Text:

The USB4 Port initiating exit from CL0s state shall:

1. Send a Low Frequency Periodic Signaling (LFPS) burst on Lane 0. The LFPS duration shall be at least 16 LFPS cycles (see Section 3.4).
 - a. If the receiver detects a CL0s_EXIT Ordered Set with the *CL0s Phase* field set to 00t, it shall continue to step 2.
 - b. If the receiver detects LFPS after receiving Gen 4 CL_OFF Ordered Sets, it shall perform the CL1/2 exit flow (starting from Step 1), which is described in Section 4.2.1.6.5.4.
 - ~~c. If the Enable Gen 4 Link Recovery bit in PORT_CS_19 is set to 1b and none of the above conditions are true after tTrainingError time after sending the first LFPS, the Port shall initiate Gen 4 Link Recovery as described in Section 4.4.7.~~
 - ~~d. If the Enable Gen 4 Link Recovery bit in PORT_CS_19 is set to 0b and none of the above conditions are true after tTrainingAbort2 time after sending the first LFPS, the Port~~

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~~shall initiate a Disconnect by driving SBTX to a logical low state for tDisconnectTx time.~~

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If the Enable Gen 4 Link Recovery bit in PORT_CS_19 is set to 1b and the initiating Port does not transition to CL0 state within tTrainingError time after sending the first LFPS, the Port shall initiate a Gen 4 Link Recovery as described in Section 4.4.7.

If the Enable Gen 4 Link Recovery bit in PORT_CS_19 is set to 0b and the initiating Port does not transition to CL0 state within tTrainingAbort2 time after sending the first LFPS, the Port shall initiate a Disconnect by driving SBTX to a logical low state for a minimum of tDisconnectTx.

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If the Enable Gen 4 Link Recovery bit in PORT_CS_19 is set to 1b and the Responding Port does not transition to CL0 state within tTrainingError time after detecting the first LFPS, the Port shall initiate a Gen 4 Link Recovery as described in Section 4.4.7.

If the Enable Gen 4 Link Recovery bit in PORT_CS_19 is set to 0b and the Responding Port does not transition to CL0 state within tTrainingAbort2 time after detecting the first LFPS, the Port shall initiate a Disconnect by driving SBTX to a logical low state for a minimum of tDisconnectTx.

(d). Section 4.2.1.6.5.4 Gen 4 Exit Flow from CL1 or CL2 State

To Text:

If the Enable Gen 4 Link Recovery bit in PORT_CS_19 is set to 1b and the ~~receiver~~ Port did not transition to CL0 state within tTrainingError time, the Router shall initiate a Gen 4 Link Recovery as described in Section 4.4.7.

If the Enable Gen 4 Link Recovery bit in PORT_CS_19 is set to 0b and the ~~receiver Port~~ did not ~~detect LFPS after transition to CL0 state within~~ tTrainingAbort2 time, the Router shall initiate a Disconnect by driving SBTX to a logical low state for tDisconnectTx.

(e) Section 4.4.2 Error Cases and Recovery

To Text:

Table 4-66. Error Cases and Impact on Logical Layer

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Error	Event	Response	Reporting
Timeout Error (TE)	<u>Gen 2/3 - Adapter entered Training state from either CL0, CL1, or CL2 state and did not transition to CL0 state within tTrainingError after sending the first SLOS1 (in Gen 2/3) or Gen 4 TS1 (in Gen 4).</u> <u>Gen 4 - Adapter has started an exit from CLx state and did not transition to CL0 state within tTrainingError (see Sections 4.2.1.6.5.1.2 and 4.2.1.6.5.4).</u>	Gen 2/3 – Go to Training.LOCK1 sub-state. Gen 4 – Initiate Gen 4 Link Recovery (see Section 4.4.7).	An Adapter shall set the <i>TE</i> bit in the <i>Logical Layer Errors</i> field to 1b. If the <i>TE</i> bit in the <i>Logical Layer Errors Enable</i> field is 1b, the Router shall send a Notification Packet with Event Code = ERR_LINK to the Connection Manager (see Section 6.5). If the <i>TE</i> bit in the <i>Logical Layer Errors Enable</i> field is 0b, the Router shall not send a Notification Packet.

(f) Section 4.6 Timing Parameters

To Text:

Table 4-72. Logical Layer Timing Parameters

Parameter	Description	Min	Max	Units
tTrainingError	The time an Adapter has to transition from Training state to CL0 state before a Timeout Error is generated. The time is measured from when the Adapter either sends the first SLOS1 <u>in Training state</u> (for Gen 2/3) or TS1 starts the exit from CLx (for Gen 4) to when it transitions to CL0 state <u>(see Sections 4.2.1.6.5.1.2 and 4.2.1.6.5.4).</u>	500 480	500 700	µs